

# Miles Franck

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## SUMMARY

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Software engineer with a technical foundation in Python, database systems, and systems-level programming in Rust, including concurrent and async application design. Experienced building backend architectures, debugging complex systems, and collaborating across technical and non-technical stakeholders. Brings a security-aware development mindset from SOC analyst experience, with CompTIA Security+ in progress. Looking to grow within a collaborative engineering team building reliable, well-architected software.

## CORE COMPETENCIES

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Software Design & Architecture • Concurrent / Async Programming • Database Design • Debugging & Troubleshooting • Secure SDLC • Agile / Scrum Collaboration • Technical Documentation

## TECHNICAL SKILLS

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**Languages & Databases:** Python, Rust, PostgreSQL

**Systems & Tooling:** Git, WSL, Tokio (async runtime)

**Security Tools (Familiar):** Shodan, Nmap, Splunk, Wireshark

## EXPERIENCE

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**Summer Fellow** — Carnegie Mellon University, Heinz College Summer Fellowship *2025*

- Completed coursework in networking fundamentals (TCP/IP), database systems, and information security
- Partnered with an external organization on a team project, gathering stakeholder requirements and evaluating SIEM tool capabilities against defined technical criteria (log ingestion, alerting, usability, cloud support)
- Delivered a live technical presentation of findings to an audience including CMU staff and outside industry guests

**SOC Operations Analyst II** — BCR Cyber *Dec 2025*

- Participated in a live-fire cyber range exercise, identifying and tracing an active attacker's presence within a simulated enterprise environment

**SOC Operations Analyst I** — BCR Cyber *Nov 2025*

- Applied technical troubleshooting and analytical skills within security operations workflows, including log analysis, alert triage, and structured incident documentation.

**Teaching Assistant** — Allegheny College *2022 – 2025*

- Provided one-on-one technical support and debugging help to students during office hours, translating complex concepts into clear, actionable guidance.
- Reviewed student code and projects for correctness, structure, and adherence to best-practice coding standards.
- Communicated across technical and non-technical audiences to align grading criteria and project expectations with faculty.

## PROJECTS

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**Rust TCP Chat Server** *2026*

- Built a multi-client TCP chat server in Rust using the Tokio asynchronous runtime, supporting concurrent client connections and real-time message broadcasting.
- Implemented a publish-subscribe messaging system using Tokio broadcast channels to efficiently distribute messages to all connected clients.
- Designed a room management system with unique join codes, user authentication, connection limits, and participant tracking.

### **BanjoDB** *2025*

- Built a dual-language architecture separating a Rust core from a Python interface to emulate enterprise-style backend isolation.
- Implemented Argon2 password hashing with unique per-user salts as part of the application's authentication layer.

### **Automated Vulnerability Analysis Tool** *2024*

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.

### **Deepseeking-Vulns** *2025*

- Developed an AI-assisted analysis workflow integrating Shodan, Ollama, and DeepSeek to process and evaluate exposed network service data.

### **CERTIFICATIONS**

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- CompTIA Security+ — CompTIA (Expected October 2026)
- SOC Operations Analyst II — BCR Cyber (Dec 2025)
- SOC Operations Analyst I — BCR Cyber (Nov 2025)
- Network Defenses 95-884 — Carnegie Mellon University, Heinz College (2025)